

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424055
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SET 15: Product Sales Analysis

Aim

To analyze monthly product sales using a **grouped bar chart, stacked area chart, table, and Tableau dashboard** for interactive exploration of first-quarter sales data.

Dataset

Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Q1. Grouped Bar Chart — First Quarter Sales

Question

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

R Program

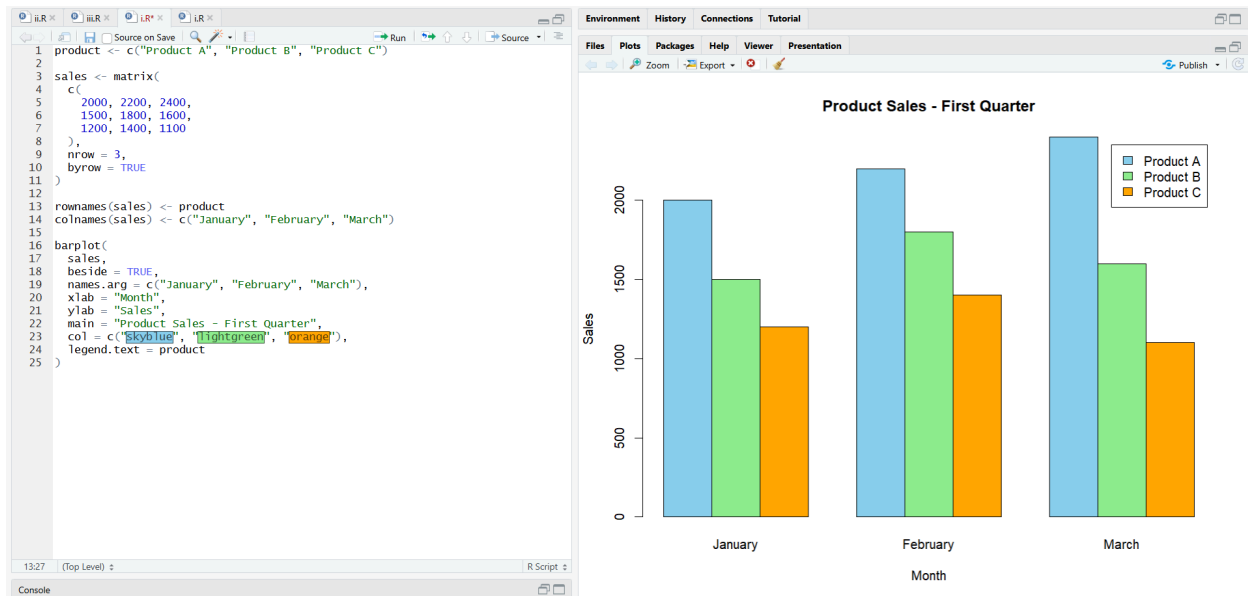
```
product <- c("Product A", "Product B", "Product C")

sales <- matrix(
  c(
    2000, 2200, 2400,
    1500, 1800, 1600,
    1200, 1400, 1100
  ),
  nrow = 3,
  byrow = TRUE
)

rownames(sales) <- product
colnames(sales) <- c("January", "February", "March")

barplot(
  sales,
  beside = TRUE,
  names.arg = c("January", "February", "March"),
  xlab = "Month",
  ylab = "Sales",
  main = "Product Sales - First Quarter",
  col = c("skyblue", "lightgreen", "orange"),
  legend.text = product
)
```

Output



Observation

- Product A has the highest sales in all three months.
- Product C has the lowest sales in all three months.
- Product A shows a continuous increase from January to March.
- Product B increases in February and decreases in March.
- Product C increases in February and decreases in March.

Q2. Stacked Area Chart — Overall Sales Trend

Question

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

R Program

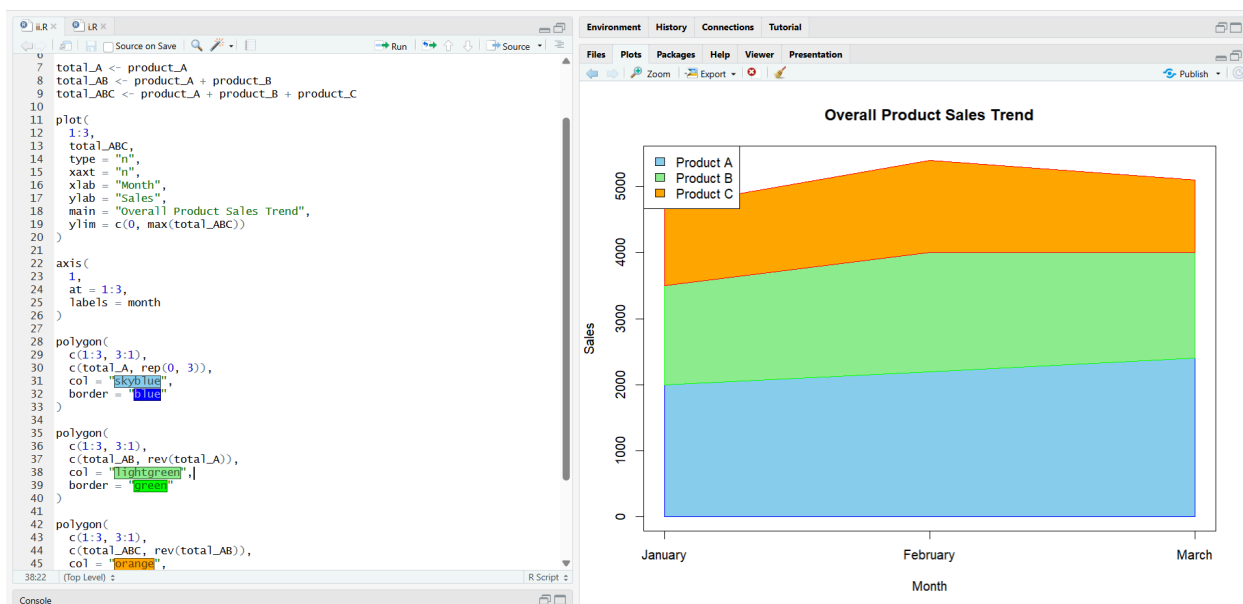
```
month <- c("January", "February", "March")
product_A <- c(2000, 2200, 2400)
product_B <- c(1500, 1800, 1600)
product_C <- c(1200, 1400, 1100)
total_A <- product_A
total_AB <- product_A + product_B
total_ABC <- product_A + product_B + product_C
plot(
  1:3,
  total_ABC,
  type = "n",
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales",
  main = "Overall Product Sales Trend",
  ylim = c(0, max(total_ABC))
)
axis(
  1,
  at = 1:3,
  labels = month
)
polygon(
  c(1:3, 3:1),
  c(total_A, rep(0, 3)),
  col = "skyblue",
  border = "blue"
)
polygon(
  c(1:3, 3:1),
  c(total_AB, rev(total_A)),
  col = "lightgreen",
```

```

    border = "green"
  )
  polygon(
    c(1:3, 3:1),
    c(total_ABC, rev(total_AB)),
    col = "orange",
    border = "red"
  )
  legend(
    "topleft",
    legend = c("Product A", "Product B", "Product C"),
    fill = c("skyblue", "lightgreen", "orange")
  )

```

Output



Observation

- Total sales increased from **4,700 in January** to **5,400 in February**.
- Total sales decreased slightly to **5,100 in March**.
- Product A contributes the largest portion of total sales throughout the quarter.

Q3. Monthly Sales Table

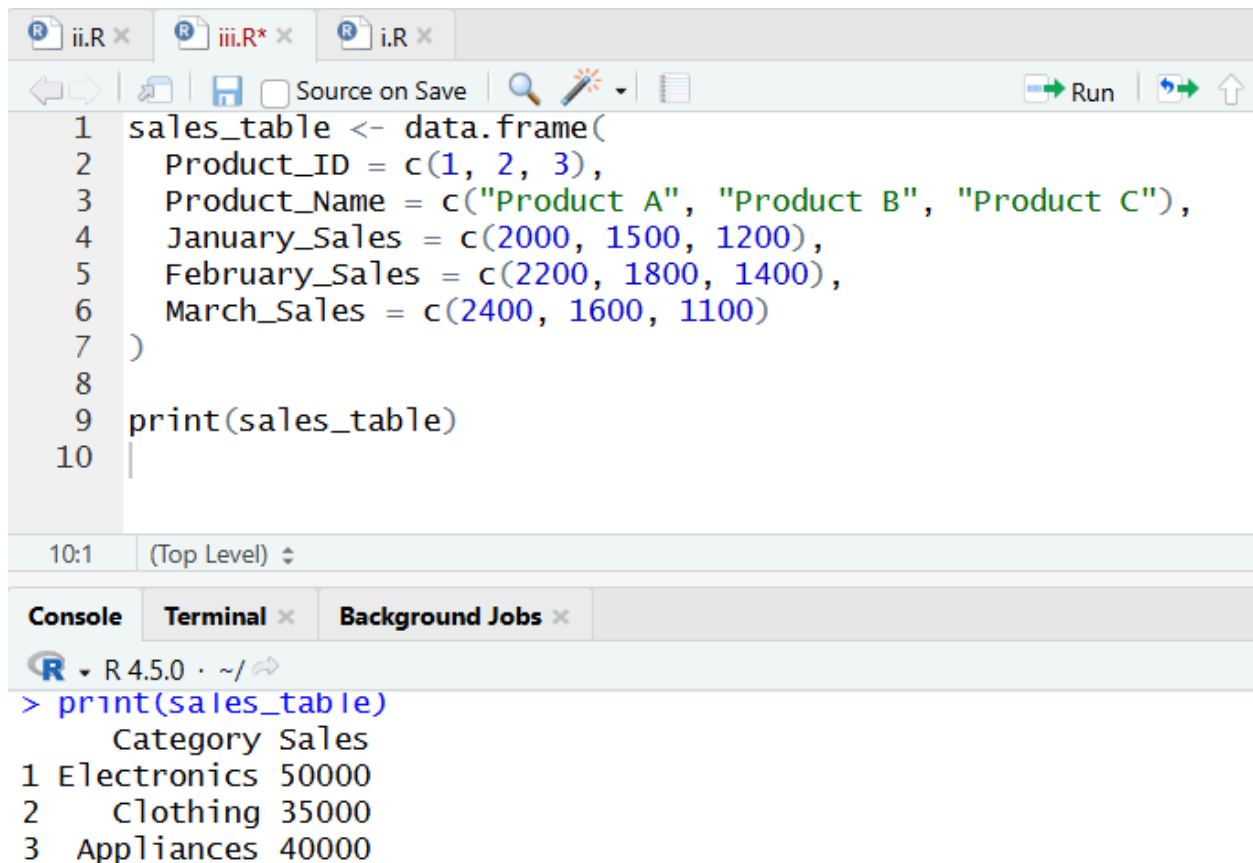
Question

Build a table to show the monthly sales data for each product. Label the table elements.

R Program

```
sales_table <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Product_Name = c("Product A", "Product B", "Product C"),  
  January_Sales = c(2000, 1500, 1200),  
  February_Sales = c(2200, 1800, 1400),  
  March_Sales = c(2400, 1600, 1100)  
)  
  
print(sales_table)
```

Output



The screenshot shows the RStudio interface. The top pane displays the R script with line numbers 1 through 10. The script creates a data frame named 'sales_table' with columns for Product ID, Product Name, and monthly sales for January, February, and March. The bottom pane shows the console output, which displays the printed data frame.

```
1 sales_table <- data.frame(  
2   Product_ID = c(1, 2, 3),  
3   Product_Name = c("Product A", "Product B", "Product C"),  
4   January_Sales = c(2000, 1500, 1200),  
5   February_Sales = c(2200, 1800, 1400),  
6   March_Sales = c(2400, 1600, 1100)  
7 )  
8  
9 print(sales_table)  
10 |
```

10:1 (Top Level) ↕

Console Terminal × Background Jobs ×

R 4.5.0 · ~/

```
> print(sales_table)  
  Category Sales  
1 Electronics 50000  
2  Clothing 35000  
3  Appliances 40000
```

Observation

The table successfully displays the monthly sales information for all three products.

Q4. Tableau Dashboard

Question

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and table for interactive exploration of sales data.

Step 1 — Prepare the Data

Create an Excel/CSV file:

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Step 2 — Import into Tableau

1. Open **Tableau**.
2. Select **Microsoft Excel** or **Text File**.
3. Import the dataset.
4. In the **Data Source** page, select the January, February, and March sales columns.
5. Use **Pivot** to convert the three month columns into rows.
6. Rename:
 - **Pivot Field Names** → **Month**
 - **Pivot Field Values** → **Sales**

This makes the data easier to use for both the bar and area charts.

Step 3 — Create Grouped Bar Chart

1. Create a new worksheet.
2. Drag **Month** → Columns.
3. Drag **Sales** → Rows.
4. Drag **Product Name** → Color.
5. Select **Bar** from the Marks card.

6. Keep the bars side-by-side.

Title: Product Sales - First Quarter

Step 4 — Create Stacked Area Chart

1. Create another worksheet.
2. Drag **Month** → Columns.
3. Drag **Sales** → Rows.
4. Change Marks type to **Area**.
5. Drag **Product Name** → Color.
6. Make sure the areas are stacked.

Title: Overall Product Sales Trend

Step 5 — Create Sales Table

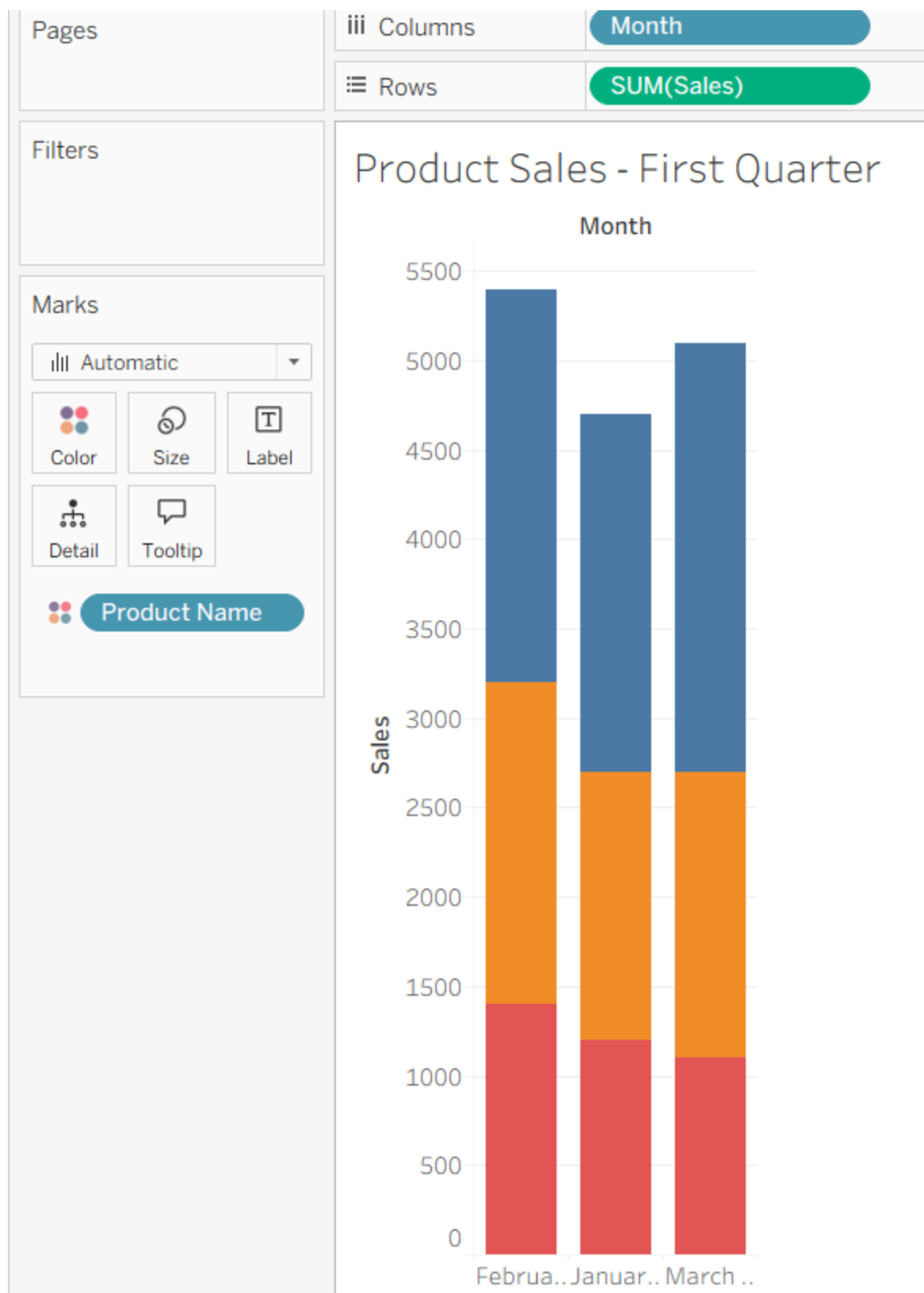
1. Create another worksheet.
2. Drag **Product Name** → Rows.
3. Drag **Month** → Columns.
4. Drag **Sales** → Text.
5. Format the values appropriately.

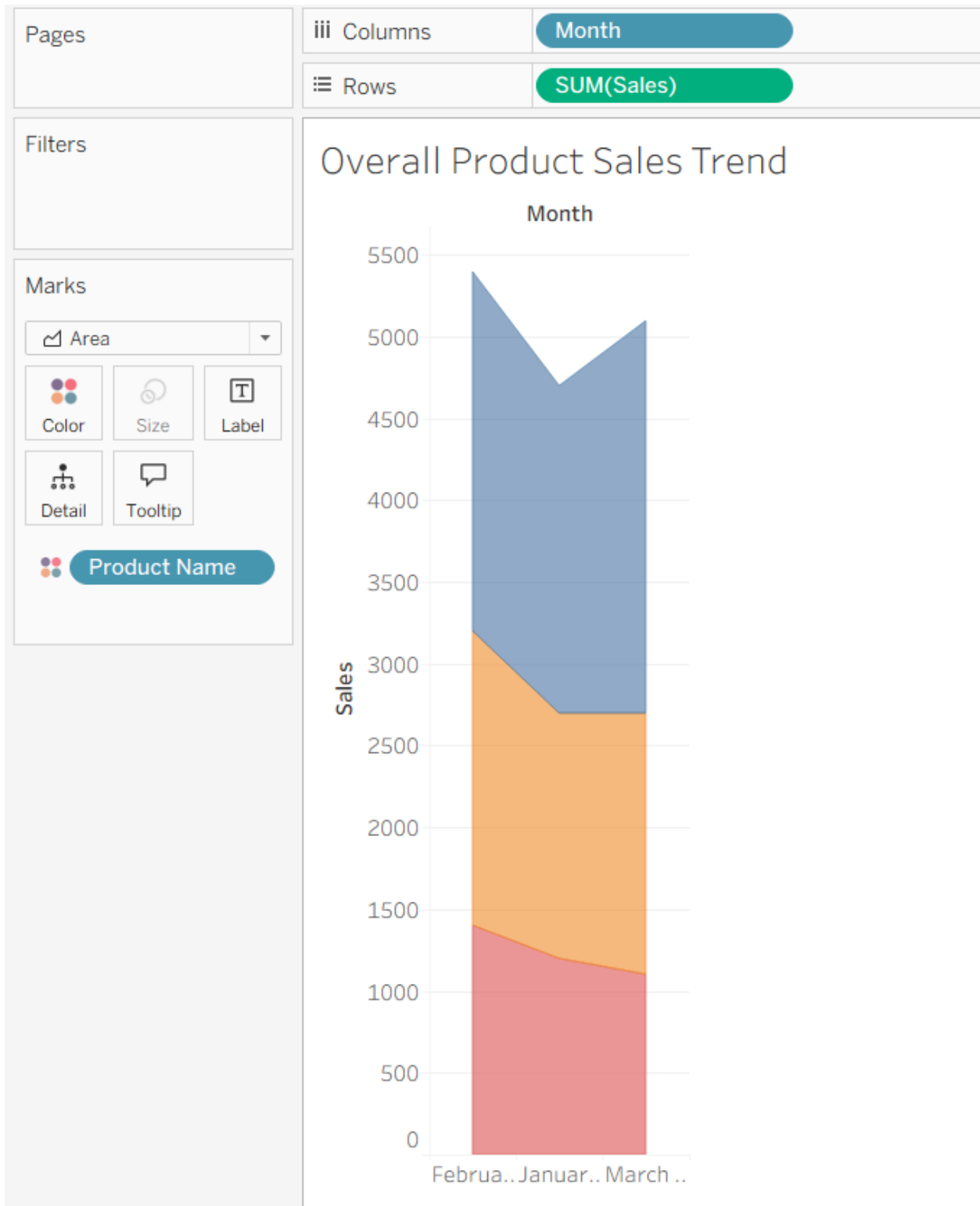
Title: Monthly Product Sales

Step 6 — Create Dashboard

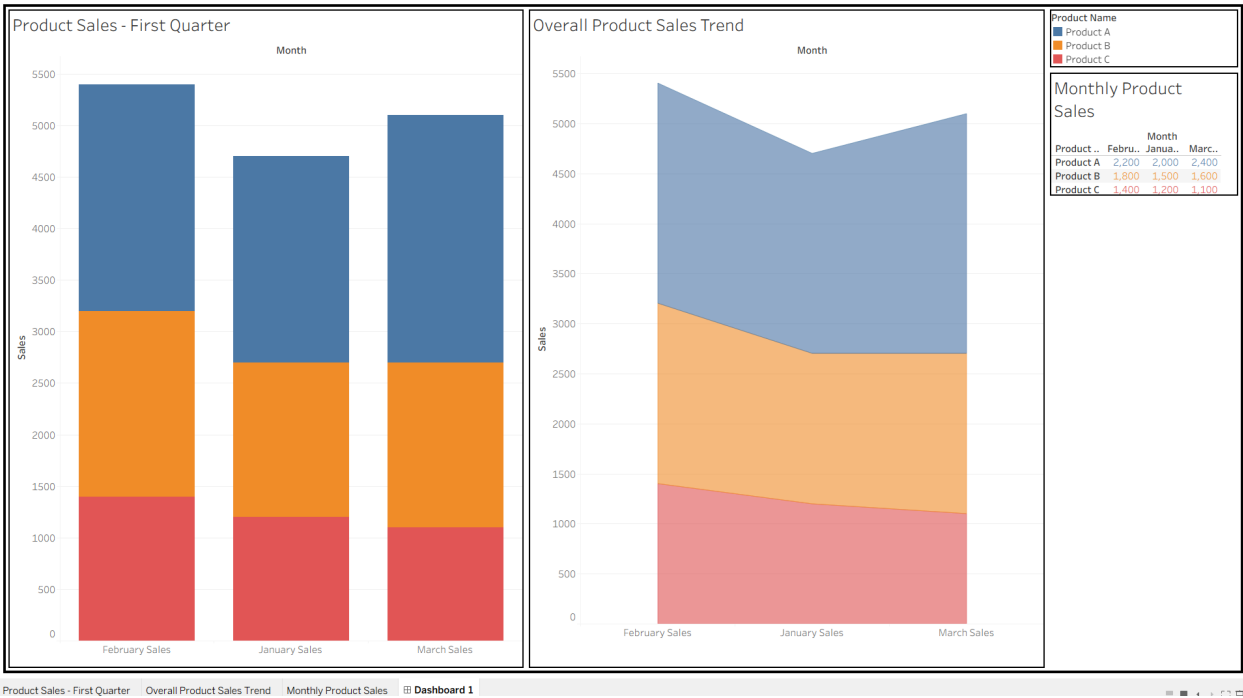
1. Select **New Dashboard**.
2. Add the **Grouped Bar Chart**.
3. Add the **Stacked Area Chart**.
4. Add the **Monthly Sales Table**.
5. Add filters for **Product Name** and **Month** if required.
6. Arrange and format the dashboard.

Expected Dashboard





Columns	Month
Rows	Product Name
Monthly Product Sales	
Month	
Product ..	Febru.. Janua.. Marc..
Product A	2,200 2,000 2,400
Product B	1,800 1,500 1,600
Product C	1,400 1,200 1,100



Procedure

1. Open **RStudio** and enter the monthly product sales dataset.
2. Create a sales matrix containing January, February, and March sales.
3. Use `barplot()` with `beside = TRUE` to create the **grouped bar chart**.
4. Calculate cumulative sales values for Products A, B, and C.
5. Use `polygon()` to create the **stacked area chart** showing the overall sales trend.
6. Use `data.frame()` to create the **monthly sales table**.
7. Open **Tableau** and import the product sales dataset.
8. Pivot the January, February, and March sales columns into a Month and Sales structure.
9. Create the grouped bar chart in Tableau.
10. Create the stacked area chart in Tableau.
11. Create a table displaying the monthly sales of each product.
12. Combine all three visualizations into a **Tableau dashboard**.
13. Add suitable filters, labels, titles, and formatting for interactive exploration.

Result

The monthly product sales data was successfully analyzed using **R and Tableau**. A grouped bar chart was created to compare first-quarter sales, a stacked area chart was created to visualize the overall sales trend, and a table was created to display monthly sales for each product. These visualizations were successfully combined into an **interactive Tableau dashboard**.